

Foundations of Reinforcement learning human feedback

[Reinforcement learning human feedback]

1.0 Content Block

where K is the number of responses the labelers ranked, $r_{\theta}(x, y)$ is the output of the reward model for prompt x and completion y , y_w is the preferred completion over y_l , $\sigma(x)$ denotes the sigmoid function, and $E[X]$ denotes the expected value. This can be thought of as a form of logistic regression, where the model predicts the probability that a response y_w is preferred over y_l . This loss function essentially measures the difference between the reward model's predictions and the decisions made by humans. The goal is to make the model's guesses as close as possible to the humans' preferences by minimizing the difference measured by this equation. In the case of only pairwise comparisons, $K = 2$, so the factor of $1 / \binom{K}{2} = 1$. In general, all $\binom{K}{2}$ comparisons from each prompt are used for training as a single batch. After training, the outputs of the model are normalized such that the reference completions have a mean score of 0. That is, $\sum_y r_{\theta}(x, y) = 0$ for each query and reference pair (x, y) by calculating the mean reward across the training dataset and setting it as the bias in the reward head.

2.0 Content Block

Next, invert this relationship to express the reward implicitly in terms of the optimal policy: $r^*(x, y) = \beta \log \frac{\pi^*(y|x) \pi_{\text{SFT}}(y|x)}{Z(x)}$.

3.0 Content Block

Finally, plug it back to the maximum likelihood estimator, we obtain $\pi^* = \arg \max_{\pi} \mathbb{E}_{(x, y_1, \dots, y_N) \sim D} [\ln \prod_{k=1}^N e^{\beta \log \frac{\pi(y_k|x) \pi_{\text{SFT}}(y_k|x)}{Z(x)}}] \sim D \left[\ln \prod_{k=1}^N \frac{e^{\beta \log \frac{\pi(y_k|x) \pi_{\text{SFT}}(y_k|x)}{Z(x)}}}{Z(x)} \right]$

4.0 Content Block

DPO eliminates the need for a separate reward model or reinforcement learning loop, treating alignment as a supervised learning problem over preference data. This is simpler to implement and train than RLHF and has been shown to produce comparable and sometimes superior results. Nevertheless, RLHF has also been shown to beat DPO on some datasets, for example, on benchmarks that attempt to measure truthfulness. Therefore, the choice of method may vary depending on the features of the human preference data and the nature of the task.

5.0 Content Block

Initialize the policy π_{ϕ}^{RL} to π_{SFT} , the policy output from SFT. Loop for many steps. Initialize a new empty dataset $D_{\pi_{\phi}^{RL}}$. Loop for many steps Sample a random prompt x from D_{RL} . Generate a response y from the policy π_{ϕ}^{RL} . Calculate the reward signal $r_{\theta}(x, y)$ from the reward model r_{θ} . Add the triple $(x, y, r_{\theta}(x, y))$ to $D_{\pi_{\phi}^{RL}}$. Update ϕ by a policy gradient method to increase

the objective function
$$\text{objective}(\phi) = \mathbb{E}_{(x,y) \sim D} \pi_{\phi}^{\text{RL}} [r_{\theta}(x,y) - \beta \log \frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)}]$$

6.0 Content Block

However, instead of doing the intermediate step of the reward model, DPO directly optimizes for the final policy. First, solve directly for the optimal policy, which can be done by Lagrange multipliers, as usual in statistical mechanics:
$$\pi^*(y|x) = \frac{\pi^{\text{SFT}}(y|x) \exp(r^*(x,y)/\beta)}{Z(x)},$$

7.0 Content Block

Identity preference optimization Identity preference optimization (IPO) is a modification to the original DPO objective that introduces a regularization term to reduce the chance of overfitting even when preference data is noisy. To solve this objective, IPO minimizes the quadratic loss function
$$\mathbb{E}_{(x,y_w,y_l) \sim D} [h_{\pi}(x,y_w,y_l) - \frac{1}{2\beta} - 1]^2$$
 where
$$h_{\pi}(x,y_w,y_l) = \log \frac{\pi_{\theta}(y_w|x) \pi_{\text{ref}}(y_l|x)}{\pi_{\theta}(y_l|x) \pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_{\theta}(y_l|x) \pi_{\text{ref}}(y_l|x)}{\pi_{\theta}(y_w|x) \pi_{\text{ref}}(y_w|x)}.$$
 IPO can control the gap between the log-likelihood ratios of the policy model and the reference by always regularizing the solution towards the reference model. It allows learning directly from preferences without a reward modelling stage and without relying on the Bradley-Terry modelling assumption that assumes that pairwise preferences can be substituted with pointwise rewards.